

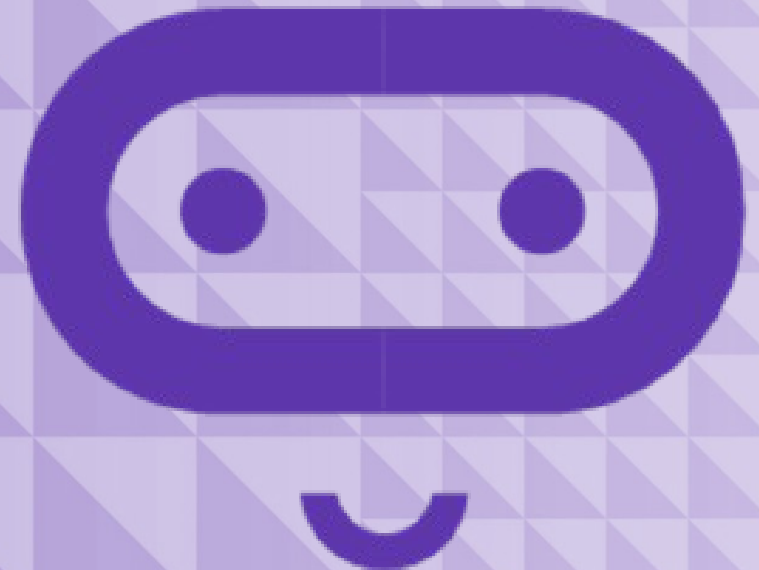
micro:bit

Introduction to Computer Science

Unit 12: Accelerometer

Lesson A

Understanding the accelerometer



What we'll learn in Lesson A

- What acceleration and velocity measure
- The accelerometer functionality of the micro:bit

Velocity

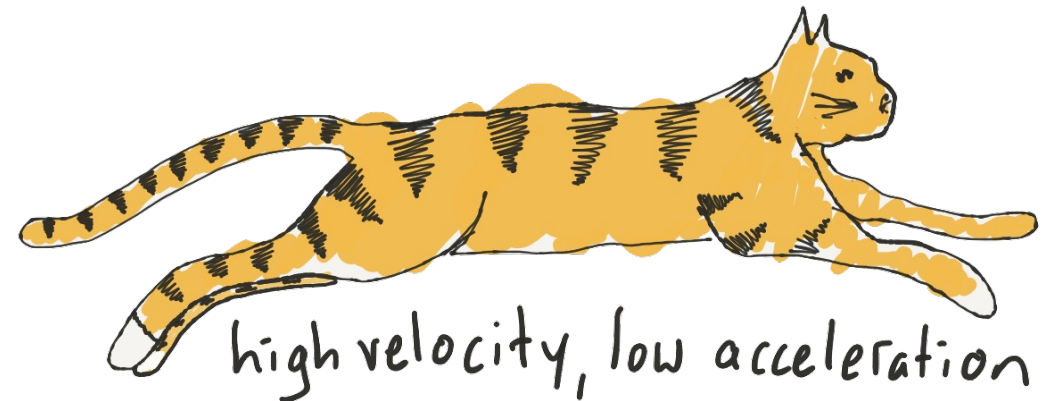
- **Velocity** describes an **object's current speed and direction**.
- Velocity is expressed as a **rate of change of position over time**.



Acceleration

Acceleration describes a **change in an object's velocity**.

A running cat that is neither slowing down nor speeding up has **zero acceleration**.



Negative acceleration

A cat that is slowing down rapidly has **high acceleration in the opposite direction** of its velocity—or **negative acceleration**.

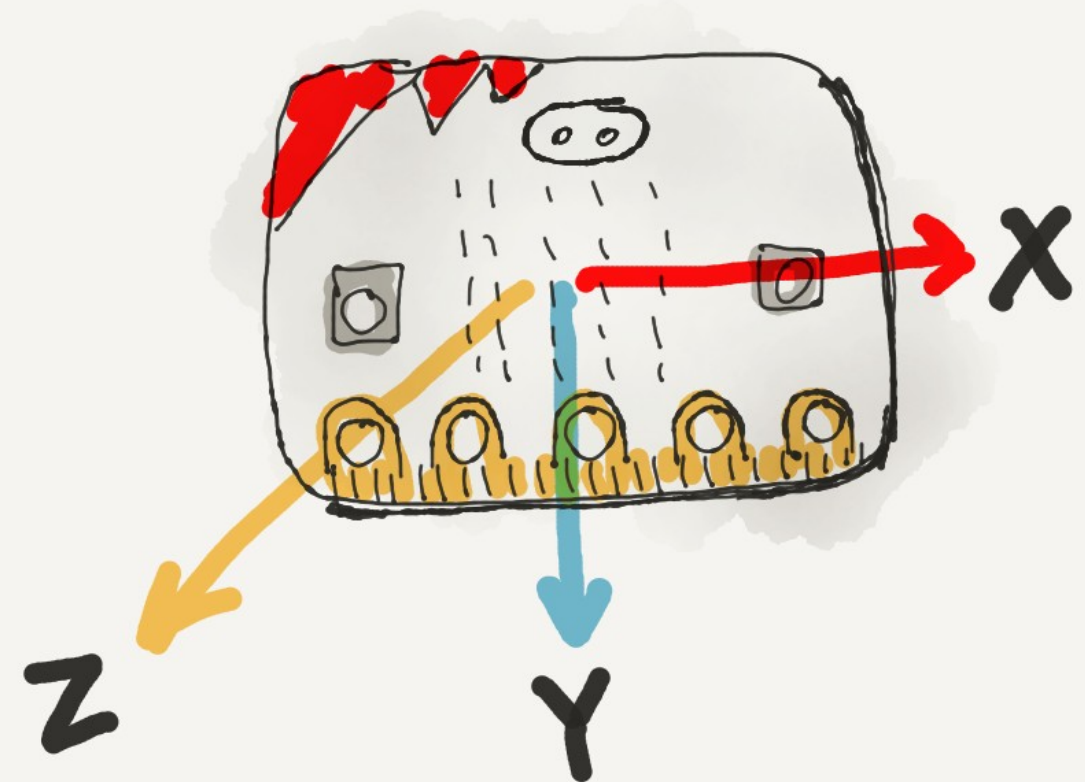


low velocity, high acceleration
(negative)

Three axes of movement

The **accelerometer** detects movement along **three separate axes of movement**

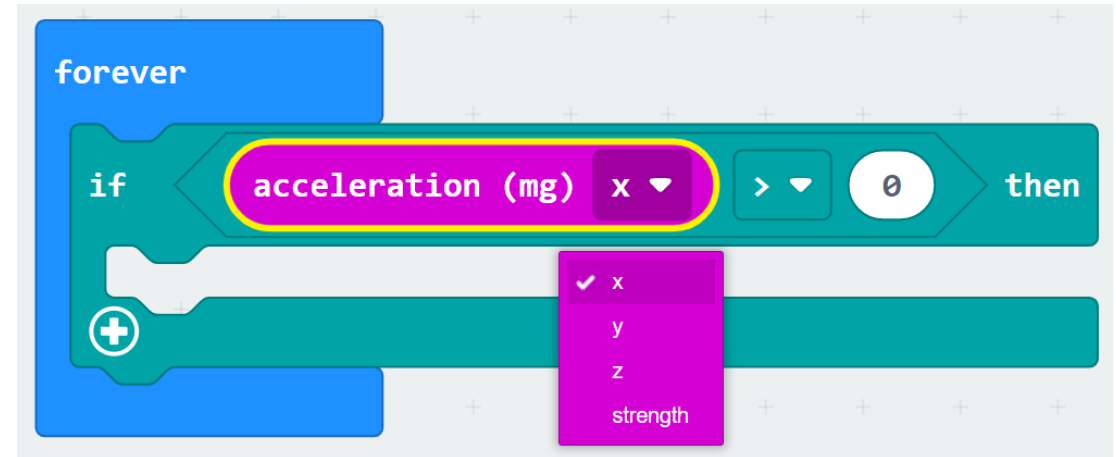
- **x-axis:** Tilt left/right
- **y-axis:** Tilt forward/back
- **z-axis:** Move up/down



face up on table

MakeCode blocks

- Use a **logic block** to see if the acceleration exceeds a certain value.
- **Select x-, y-, or z-axis** from the **dropdown menu**.



Strike a pose

- One student chooses a pose.
- Everybody else imitates that same pose.
- How does your body maintain its balance?



What we learned in Lesson A



- Acceleration measures a change in velocity
- The micro:bit measures acceleration along three axes of movement
- Next time: Checking for acceleration using code

Lesson B

Explore the accelerometer tools

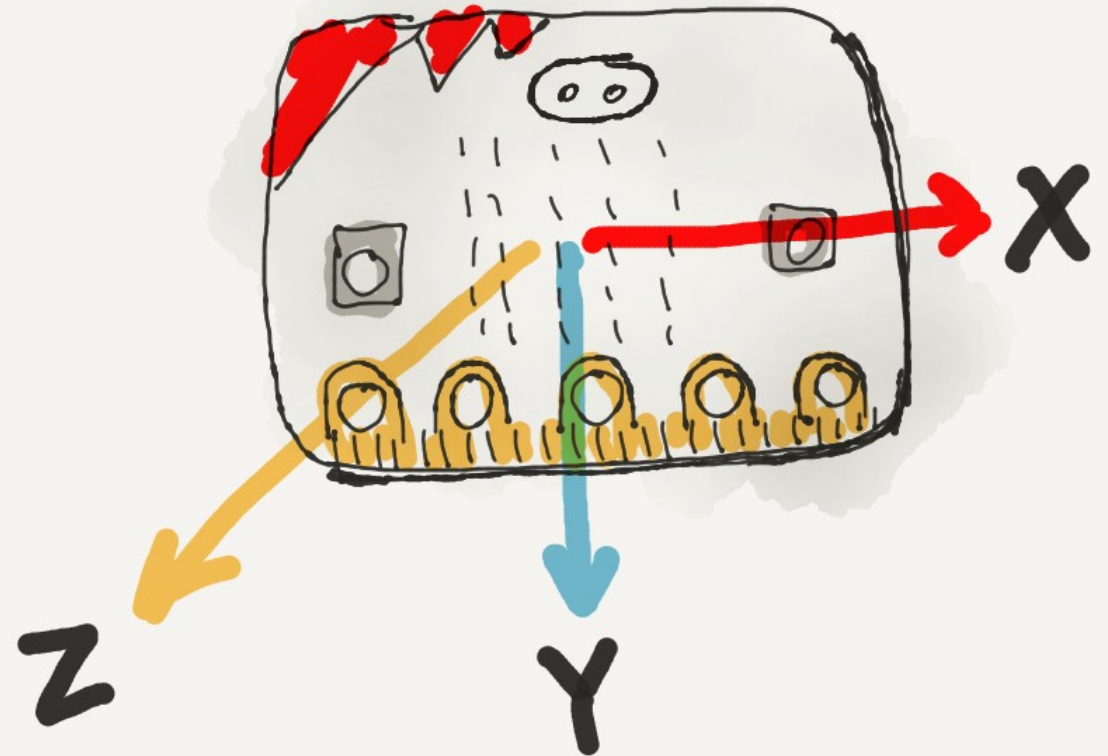


What we'll learn in Lesson B

- How to detect motion using the micro:bit

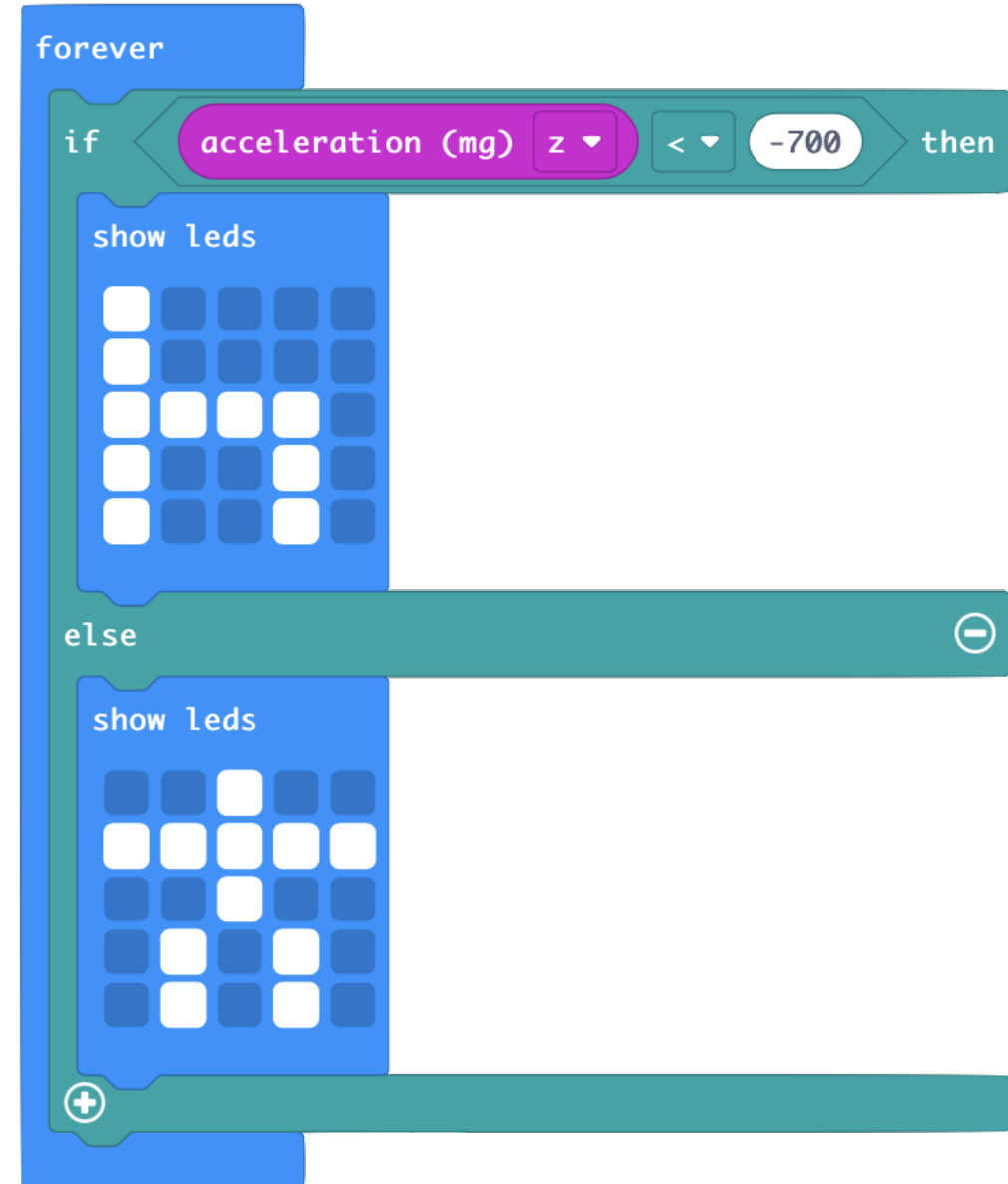
Stand for health!

In this activity, you will create a micro:bit program that will remind you when it's time to stand up and take a break.



Write your code

1. Follow the steps in the **Student Workbook** to first check if the person is standing and then display the appropriate icon.
2. Then, write code to track how long someone has been sitting down.



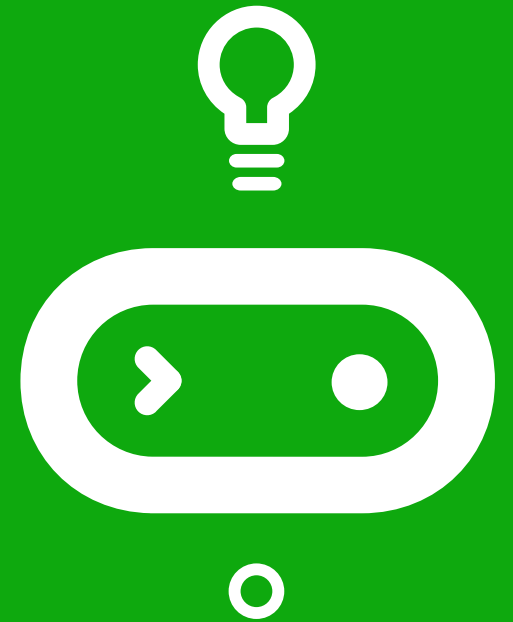
Fasten your micro:bit

You will need to tape the micro:bit above your knee for it to work.

When you stand up, it should change to a person standing.



Time for a quiz!



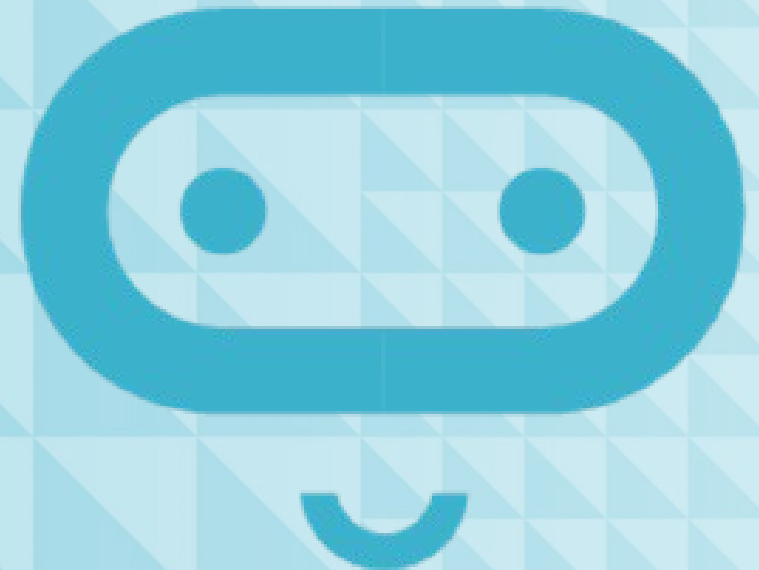
What we learned in Lesson B

- Coded a health app
- Created a physical holder to securely attach the micro:bit to a person
- Tested and made adjustments



Lesson C

Make an accelerometer project



What we'll learn in Lesson C

- Create a micro:bit project that uses the accelerometer and other features to accomplish a mission
- Learn about the UN Global Goals for Sustainability

1 mission, 2 features

Pick a mission from the list:

- Find a lost item
- Measure something
- Make friends
- Teach or learn something
- Make up your own!

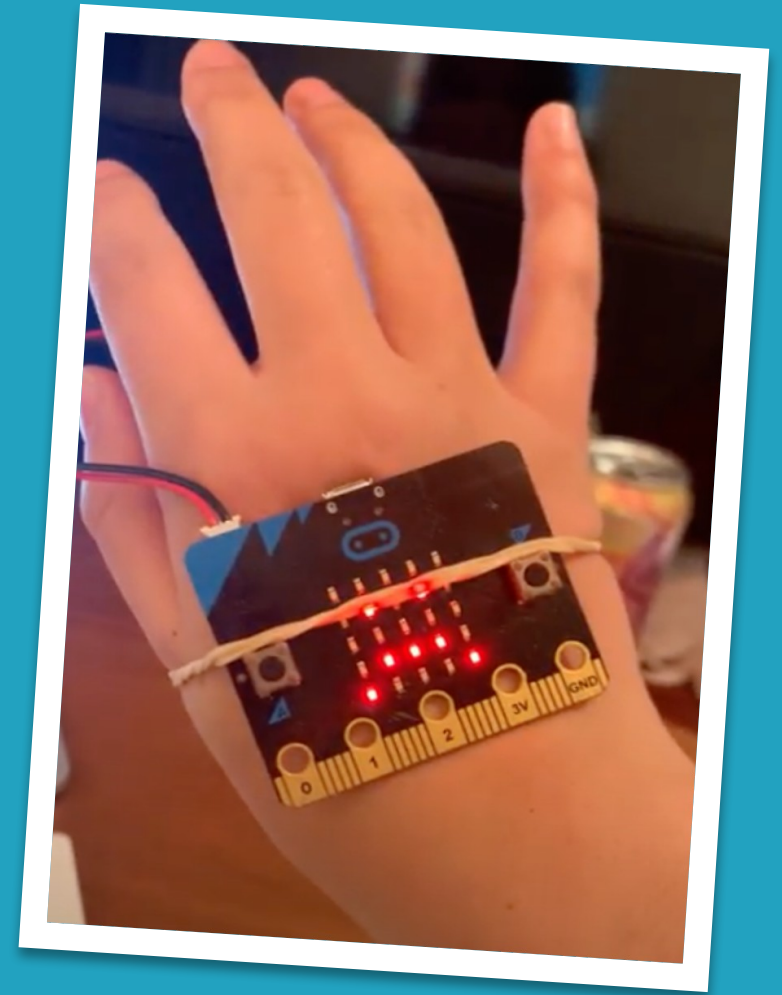
Feature #1: Accelerometer

Choose a Feature #2:

- | | |
|---------------|----------------------|
| ▪ Button A | ▪ Shake |
| ▪ Button B | ▪ Temperature sensor |
| ▪ Buttons A+B | ▪ Radio |
| ▪ Compass | ▪ Light sensor |

Example: Don't touch your face!

Diseases spread when people touch contaminated keyboards and then touch their face. This project teaches people to stay healthy while working on computers. The accelerometer detects when the student raises their hand off the keyboard and displays a frowny face.



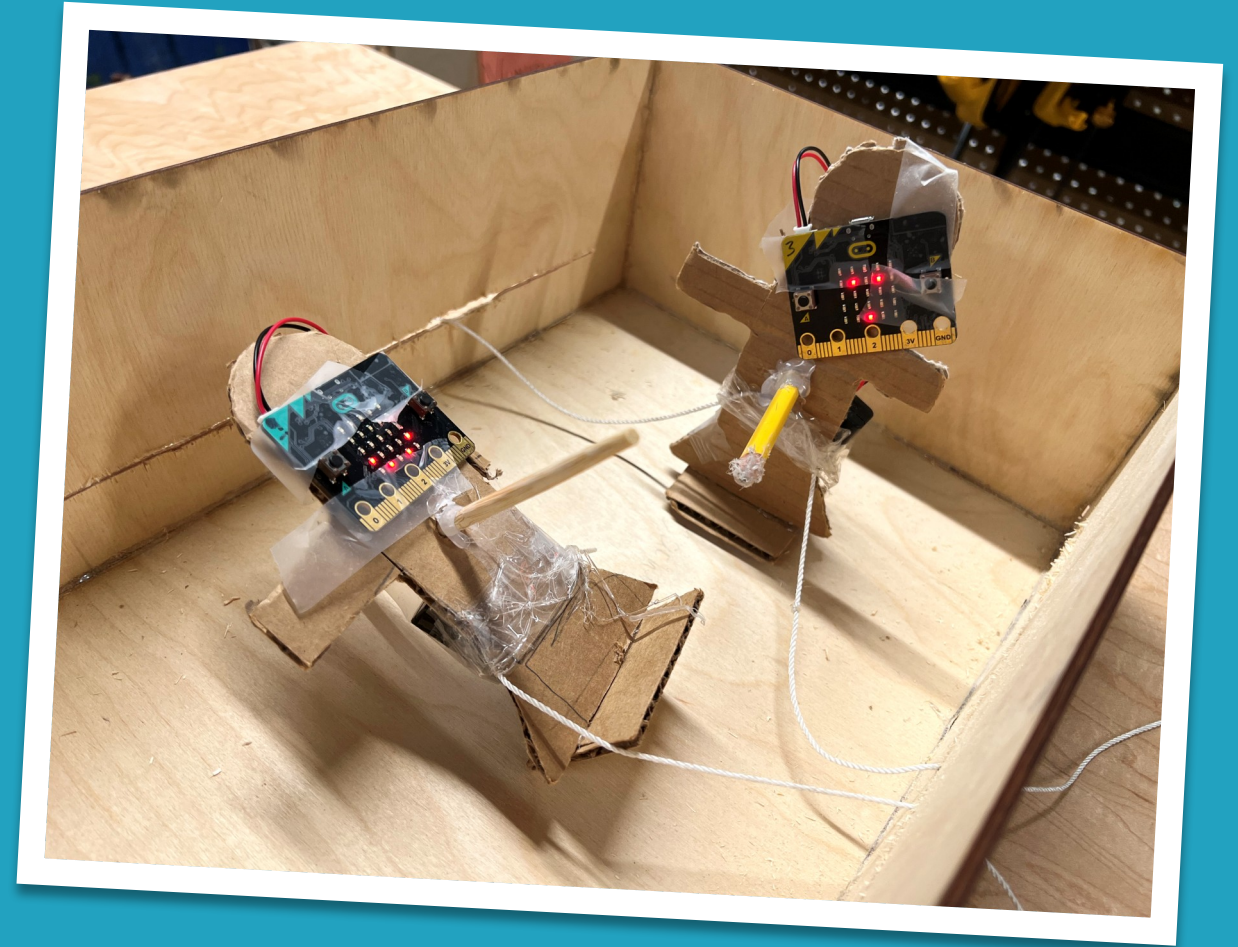
Example: Pedometer

Just being aware of how many steps you take during the day can help you stay active. This project uses the accelerometer to detect when the wearer takes a step and saves the total amount of steps it counts.



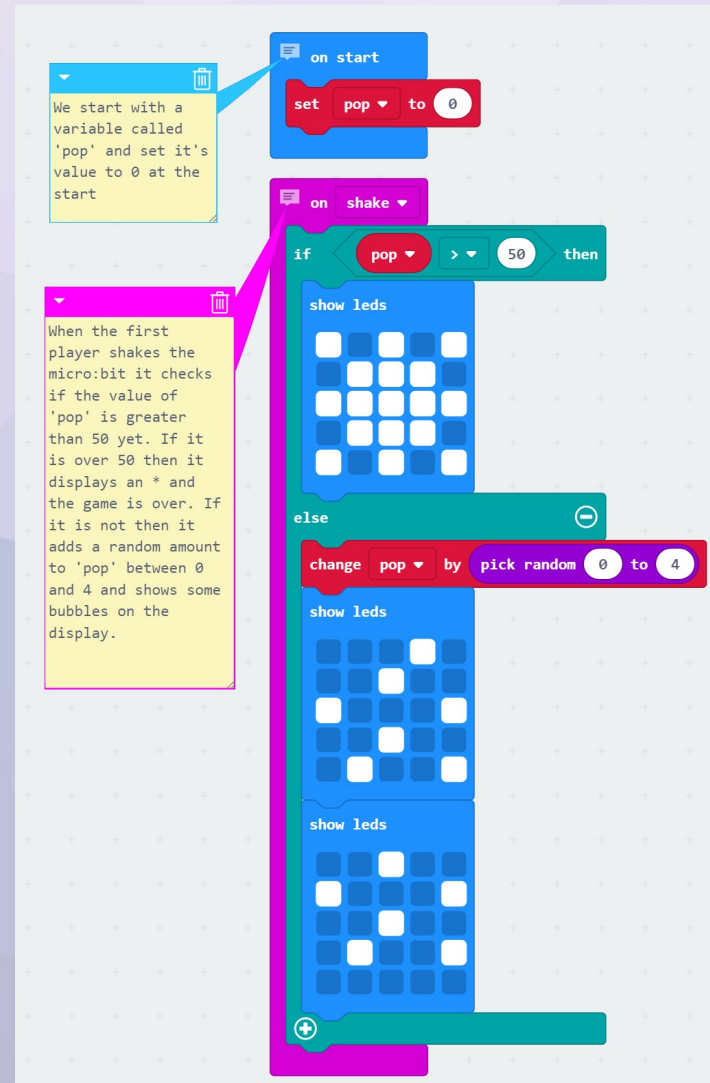
Example: Fencing

This project teaches how points are scored in Olympic fencing. Each person has a micro:bit whose accelerometer detects if it has been hit hard enough with the foil to score a point.

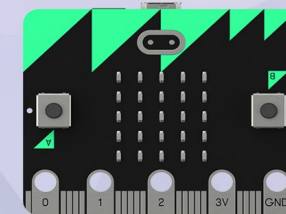




SHAKE THE BOTTLE



This game is also inspired by a game in the Nintendo Switch game 12Switch. Players take it in turn to hold the micro:bit, which could be taped to a bottle. They give it a shake and some bubbles show on the screen. They pass it to the next player who also gives it a shake. At some point the bottle will pop and the player left holding the bottle loses.



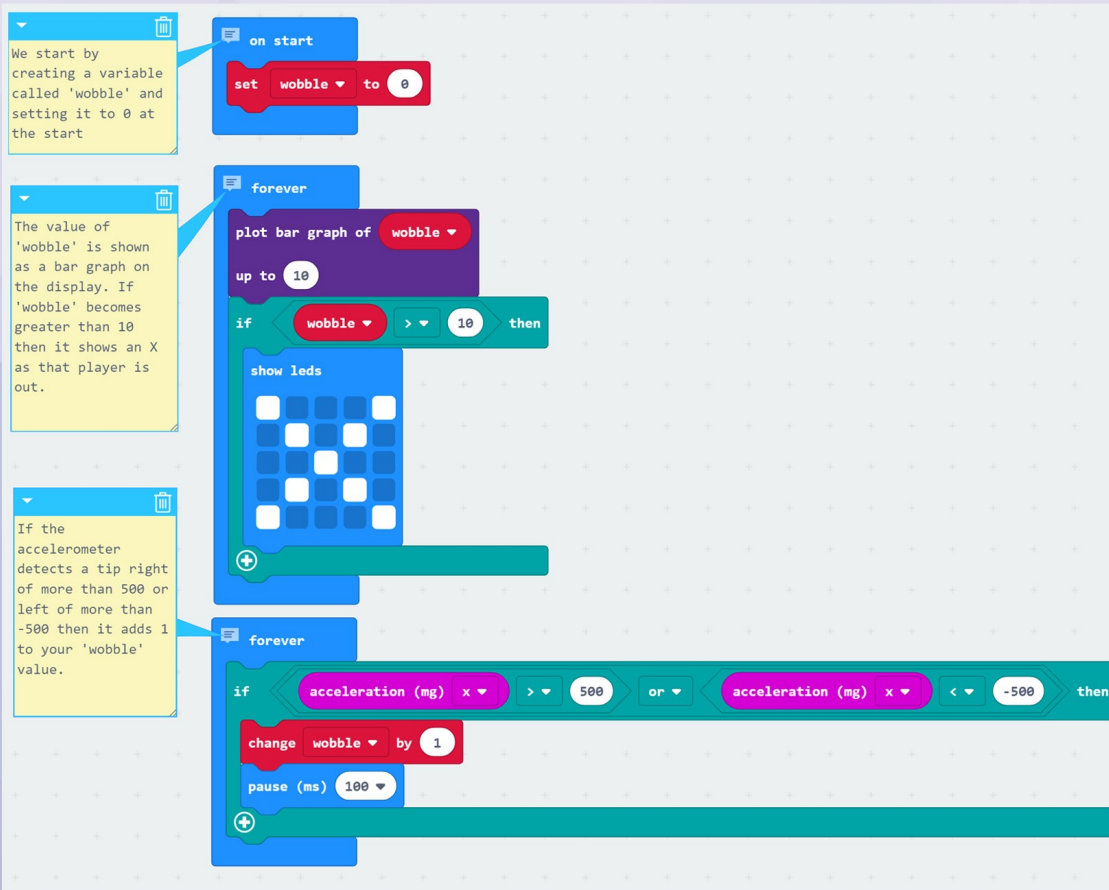
There are no extra components needed for this game as it is just using the built in accelerometer to detect when the bottle is shaken.

MAKING MICRO:BIT GAMES

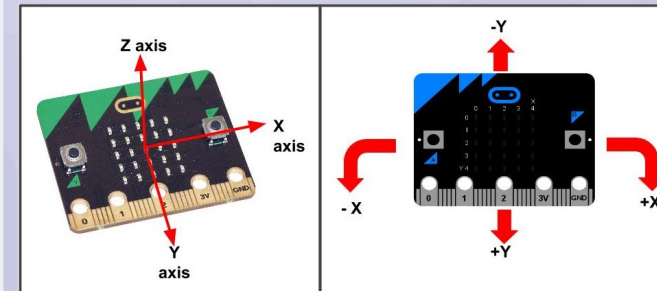


ZEN

ZEN is a game our students made that was inspired by the Nintendo Switch game 12Switch. The aim of the game is to strike a yoga pose and hold it while keeping the Micro:bit completely still. If you wobble your screen starts to fill up. If it is full then you are out.



There are no extra components needed for this game as it is just using the built in accelerometer on the x axis to detect how much it wobbles rotationally left and right. You could tape the micro:bit and battery to a piece of card to make it easier to hold flat.



Project expectations

Use a design thinking approach and ensure the project meets these specifications:

- Uses the accelerometer to gather input about the micro:bit's position and motion in space
- Uses accelerometer blocks in a way that is integral to the program
- The program compiles and runs as intended and includes meaningful comments in the code
- Provide the written Reflection Diary entry

Reflection Diary

- What kind of project did you do? What was your mission/feature?
- How does your project use the accelerometer?
- Describe something in your project that you are proud of.
- Describe a difficult point in the process of designing this program and explain how you resolved it.
- What feedback did your testers give you?
How did that help you improve your design?
- How would you improve your project, given more time?
- Publish your MakeCode program and include the link.

